

Object Oriented Analysis and Designs ICT 238-3

Lab Sheet 01

1. Write a Java program to display your name, age, university, and GPA using appropriate data types.
2. Write a Java program to input the length and width of a rectangle and calculate its area and perimeter.
3. Write a Java program to input the radius of a circle and calculate its area and circumference. (Use $\pi = 3.14159$)
4. Write a Java program to input two integers and display
 - a. Sum
 - b. Difference
 - c. Product
 - d. Quotient
 - e. Remainder
5. Write a Java program to input three numbers and calculate
 - a. Total
 - b. Average
6. Write a Java program to input a number and determine whether it is even or odd using the modulus (%) operator.
7. Write a Java program to input a student's marks and display **"Pass"** if the marks are 50 or above; otherwise display **"Fail"**.
8. Write a Java program to input two numbers and display the larger number.
9. Write a Java program to input three numbers and display the largest number.
10. Write a Java program to input a student's marks and display the grade according to the following criteria

Marks	Grade
75–100	A
65–74	B
50–64	C
Below 50	F

11. Write a Java program to input a month number (1-12) and display the corresponding month name using a switch statement.
12. Write a Java program to display the numbers from 1 to 100 using a for loop.
13. Write a Java program to display all even numbers between 1 and 50.
14. Write a Java program to calculate the sum of all numbers from 1 to N, where N is entered by the user.
15. Write a Java program to display the multiplication table of a number entered by the user.

Example:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

...

$$5 \times 10 = 50$$

16. Write a Java program to calculate the factorial of a positive integer entered by the user.
17. Write a Java program to count the number of digits in an integer.
Input: 45678
Output: 5
18. Write a Java program to reverse a given integer.
Input: 12345
Output: 54321
19. Write a Java program to input 10 integers and determine
 - a. The sum of all numbers
 - b. The average
 - c. The largest number
 - d. The smallest number
20. Write a Java program to input N numbers and count how many are
 - a. Even
 - b. Odd
 - c. Positive
 - d. Negative
21. Write a Java program to input marks for 5 students and display the grade (A, B, C, or F) for each student based on the grading criteria given in Question 10.
22. Write a Java program to repeatedly ask the user to enter a number. The program should stop when the user enters 0. After the loop ends, display
 - a. Total numbers entered (excluding 0)
 - b. Sum of the numbers
 - c. Average of the numbers
23. Write a Java program that simulates a simple ATM menu using a do-while loop and a switch statement.

Menu

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

Requirements

- Start with an initial balance of Rs. 10,000.
- Allow the user to perform multiple transactions until they choose Exit.
- Prevent withdrawals that exceed the available balance.